COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE212 - Automating Server Management	**SECTION:**	CPE31S2
DATE PERFORMED:	09/07/2026	**DATE SUBMITTED:**	30/07/2026
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2.1 HANDS-ON ACTIVITY — SSH Key-Based Authentication and GIT Setup

1. PROCEDURE OUTPUT

Task 1: Create an SSH Key Pair for User Authentication

1. The simplest way to generate a key pair is to run ssh-keygen without arguments. In this case, it will prompt for the file in which to store keys. First, the tool asked where to save the file. SSH keys for user authentication are usually stored in the users .ssh directory under the home directory. However, in enterprise environments, the location is often different. The default key file name depends on the algorithm, in this case id_rsa when using the default RSA algorithm. It could also be, for example, id_dsa or id_ecdsa.
2. Issue the command ssh-keygen -t rsa -b 4096. The algorithm is selected using the -t option and key size using the -b option.
3. When asked for a passphrase, just press enter. The passphrase is used for encrypting the key, so that it cannot be used even if someone obtains the private key file. The passphrase should be cryptographically strong.

```
queleroyubuntu3@workstation:~$ ssh-keygen -t rsa -b 4096
Generating public/private rsa key pair.
Enter file in which to save the key (/home/queleroyubuntu3/.ssh/id_rsa):
/home/queleroyubuntu3/.ssh/id_rsa already exists.
Overwrite (y/n)? y
Enter passphrase (empty for no passphrase):
Enter same passphrase again:
Your identification has been saved in /home/queleroyubuntu3/.ssh/id_rsa
Your public key has been saved in /home/queleroyubuntu3/.ssh/id_rsa.pub
The key fingerprint is:
SHA256:QzJv/y9mvl++MX84uwqR+ow92aJYF4G8VVb9iF17Ycg queleroyubuntu3@workstation
The key's randomart image is:
+---[RSA 4096]----+
|            +.o. |
|       . . o E oo|
|      o + o  o.o+|
|       = o o. o.o|
|        S +     .|
|       . + o     |
|        o +o   +.|
|       o *+o= oo=|
|      . o.+*=*===|
+----[SHA256]-----+
```

4. Verify that you have created the key by issuing the command ls -la .ssh. The command should show the .ssh directory containing a pair of keys. For example, id_rsa.pub and id_rsa.

```
queleroyubuntu3@workstation:~$ ls -la .ssh
total 24
drwx------  2 queleroyubuntu3 queleroyubuntu3 4096 Jul 30 09:16 .
drwxr-x--- 16 queleroyubuntu3 queleroyubuntu3 4096 Jul 16 10:24 ..
-rw-------  1 queleroyubuntu3 queleroyubuntu3 3401 Jul 30 09:18 id_rsa
-rw-r--r--  1 queleroyubuntu3 queleroyubuntu3  753 Jul 30 09:18 id_rsa.pub
-rw-------  1 queleroyubuntu3 queleroyubuntu3 2240 Jul 30 08:58 known_hosts
-rw-------  1 queleroyubuntu3 queleroyubuntu3 1120 Jul 30 08:52 known_hosts.old
```

Task 2: Copying the Public Key to the remote servers

1. To use public key authentication, the public key must be copied to a server and installed in an authorized_keys file. This can be conveniently done using the ssh-copy-id tool.
2. Issue the command similar to this: ssh-copy-id -i ~/.ssh/id_rsa user@host
3. Once the public key has been configured on the server, the server will allow any connecting user that has the private key to log in. During the login process, the client proves possession of the private key by digitally signing the key exchange.

```
queleroyubuntu3@workstation:~$ ssh-copy-id -i ~/.ssh/id_rsa queleroyubuntu3@serv
er1
/usr/bin/ssh-copy-id: INFO: Source of key(s) to be installed: "/home/queleroyubu
ntu3/.ssh/id_rsa.pub"
/usr/bin/ssh-copy-id: INFO: attempting to log in with the new key(s), to filter
out any that are already installed
/usr/bin/ssh-copy-id: INFO: 1 key(s) remain to be installed -- if you are prompt
ed now it is to install the new keys
queleroyubuntu3@server1's password:

Number of key(s) added: 1

Now try logging into the machine, with:   "ssh 'queleroyubuntu3@server1'"
and check to make sure that only the key(s) you wanted were added.
```

```
queleroyubuntu3@workstation:~$ ssh-copy-id -i ~/.ssh/id_rsa queleroyubuntu3@serv
er2
/usr/bin/ssh-copy-id: INFO: Source of key(s) to be installed: "/home/queleroyubu
ntu3/.ssh/id_rsa.pub"
/usr/bin/ssh-copy-id: INFO: attempting to log in with the new key(s), to filter
out any that are already installed
/usr/bin/ssh-copy-id: INFO: 1 key(s) remain to be installed -- if you are prompt
ed now it is to install the new keys
queleroyubuntu3@server2's password:

Number of key(s) added: 1

Now try logging into the machine, with:   "ssh 'queleroyubuntu3@server2'"
and check to make sure that only the key(s) you wanted were added.
```

4. On the local machine, verify that you can SSH with Server 1 and Server 2.

```
queleroyubuntu3@workstation:~$ ssh queleroyubuntu3@server1
Welcome to Ubuntu 22.04.5 LTS (GNU/Linux 6.8.0-136-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

Last login: Thu Jul 30 08:57:43 2026 from 192.168.56.11
```

```
queleroyubuntu3@workstation:~$ ssh queleroyubuntu3@server2
Welcome to Ubuntu 22.04.5 LTS (GNU/Linux 6.8.0-136-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

New release '24.04.4 LTS' available.
Run 'do-release-upgrade' to upgrade to it.

Last login: Thu Jul 30 08:58:41 2026 from 192.168.56.11
```

What did you notice? Did the connection ask for a password? If not, why?

- Since the SSH public key was already sent to the servers through the command ssh-copy-id. It did not ask for the password.

Reflections:

Answer the following:

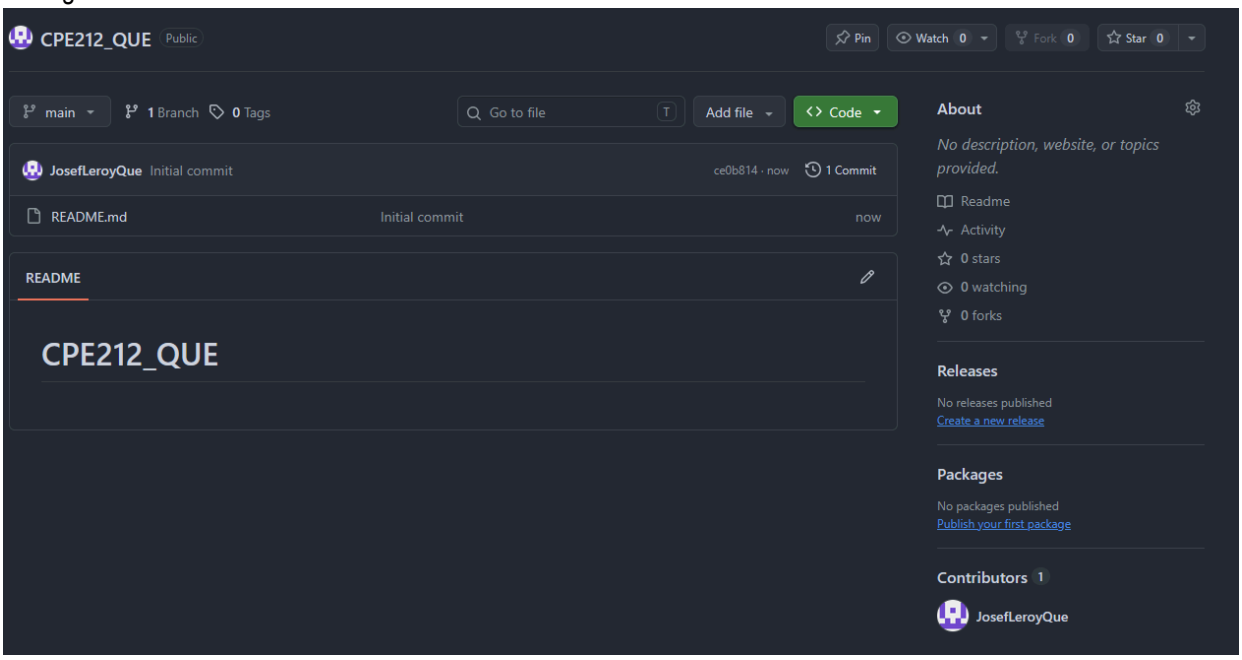
1. How will you describe the ssh-program? What does it do?

- The SSH allows us to group/connect computers over a network. This will help us secure the data we are controlling.

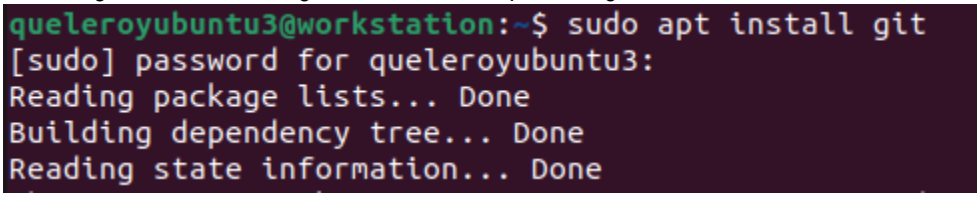
2. How do you know that you already installed the public key to the remote servers?

- Since I could already to SSH to the remote servers without the password asking, I know that it is already installed.

Part 2: Discussion
Provide screenshots for each task.
It is assumed that you are done with the last activity (Activity 2: SSH Key-Based Authentication).
Set up Git
At the heart of GitHub is an open-source version control system (VCS) called Git. Git is responsible for everything GitHub-related that happens locally on your computer. To use Git on the command line, you'll need to download, install, and configure Git on your computer. You can also install GitHub CLI to use GitHub from the command line. If you don't need to work with files locally, GitHub lets you complete many Git-related actions directly in the browser, including:
• Creating a repository
• Forking a repository
• Managing files
• Being social



Task 3: Set up the Git Repository
1. On the local machine, verify the version of your git using the command which git. If a directory of git is displayed, then you don't need to install git. Otherwise, to install git, use the following command: sudo apt install git



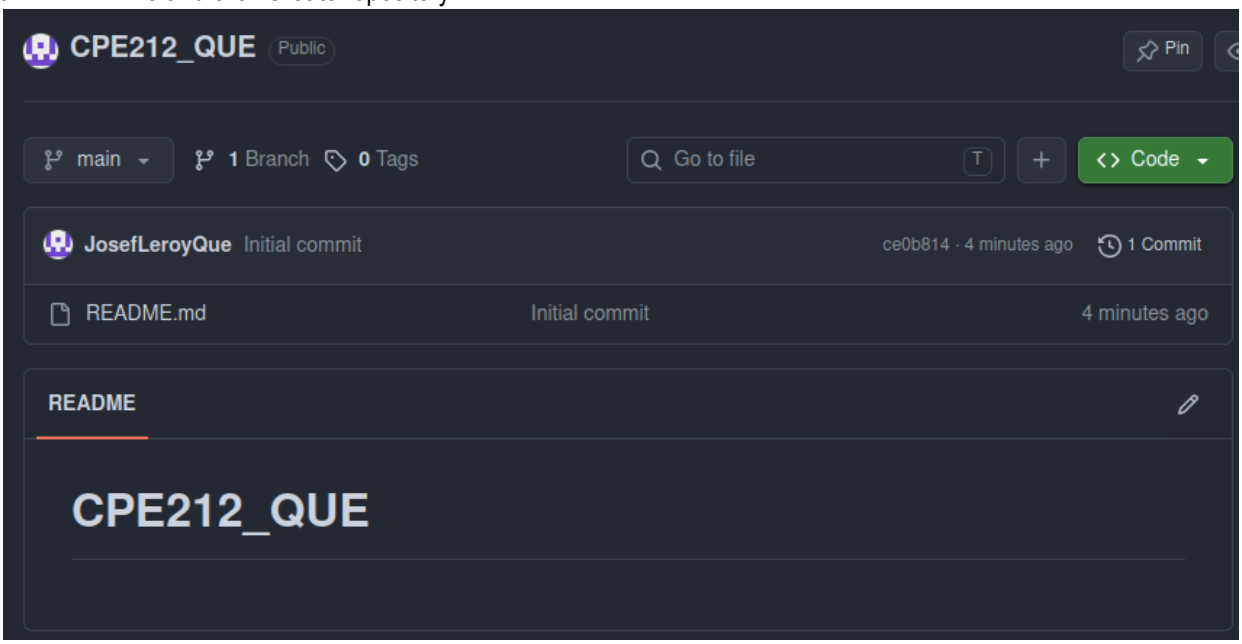
2. After the installation, issue the command which git again. The directory of git is usually installed in this location: user/bin/git.

```
queleroyubuntu3@workstation:~$ which git
/usr/bin/git
```

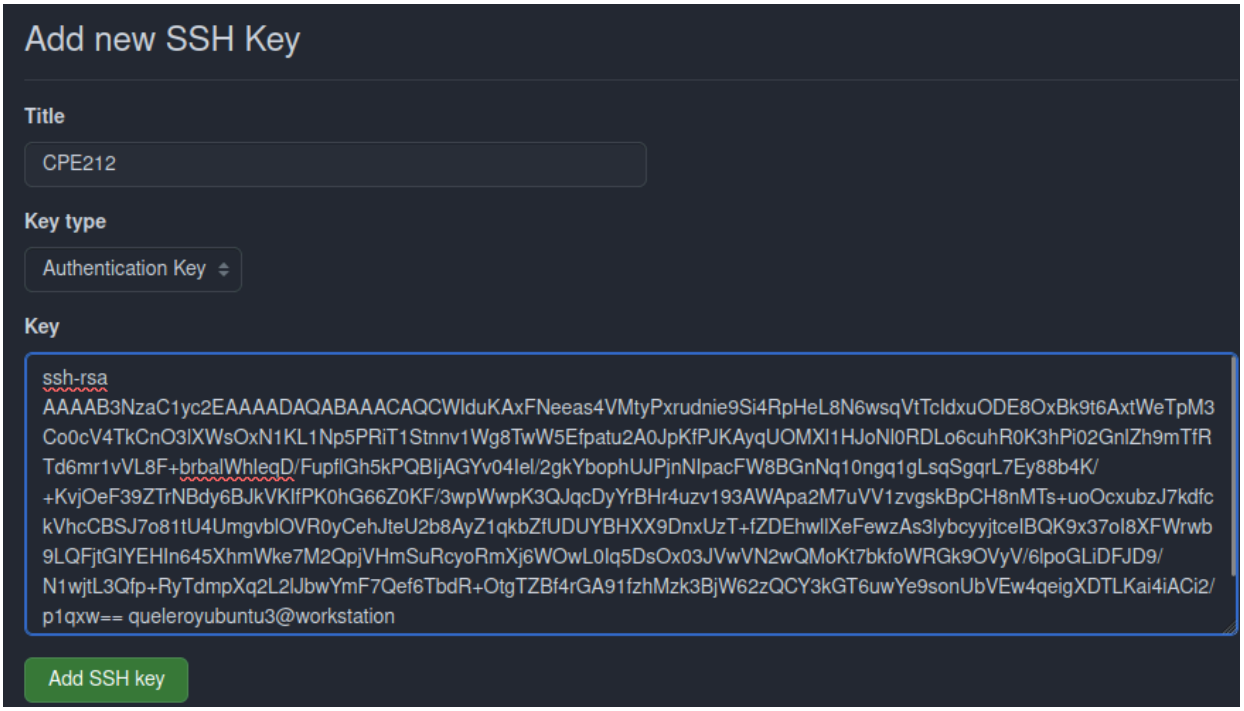
3. The version of git installed in your device is the latest. Try issuing the command git --version to know the version installed.

```
queleroyubuntu3@workstation:~$ git --version
git version 2.34.1
```

4. Using the browser in the local machine, go to www.github.com.
5. Sign up in case you don't have an account yet. Otherwise, login to your GitHub Account.
a. Create a new repository and name it as CPE232_yourname. Check Add a README file and click Create repository.



b. Create a new SSH key on GitHub. Go your profile's setting and click SSH and GPG keys. If there is an existing key, make sure to delete it. To create a new SSH keys, click New SSH Key. Write CPE232 key as the title of the key.
c. On the local machine's terminal, issue the command cat .ssh/id_rsa.pub and copy the public key. Paste it on the GitHub key and press Add SSH Key.



d. Clone the repository that you created. In doing this, you need to get the link from GitHub. Browse to your repository as shown below. Click on the Code drop down menu. Select SSH and copy the link.

e. Issue the command git clone followed by the copied link. For example, git clone git@github.com:jvtaylar-cpe/CPE232_yourname.git. When prompted to continue connecting, type yes and press enter.

```
queleroyubuntu3@workstation:~$ git clone git@github.com:JosefLeroyQue/CPE212_QUE
.git
Cloning into 'CPE212_QUE'...
The authenticity of host 'github.com (20.205.243.166)' can't be established.
ED25519 key fingerprint is SHA256:+DiY3wvvV6TuJJhbpZisF/zLDA0zPMSvHdkr4UvCOqU.
This key is not known by any other names
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added 'github.com' (ED25519) to the list of known hosts.
remote: Enumerating objects: 3, done.
remote: Counting objects: 100% (3/3), done.
remote: Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
Receiving objects: 100% (3/3), done.
```

f. To verify that you have cloned the GitHub repository, issue the command ls. Observe that you have the CPE232_yourname in the list of your directories. Use CD command to go to that directory and LS command to see the file README.md.

```
queleroyubuntu3@workstation:~$ ls
CPE212_QUE  Documents  Music     Public  Templates
Desktop     Downloads  Pictures  snap    Videos
queleroyubuntu3@workstation:~$ cd CPE212_QUE
queleroyubuntu3@workstation:~/CPE212_QUE$ ls
README.md
queleroyubuntu3@workstation:~/CPE212_QUE$
```

g. Use the following commands to personalize your git.
• git config --global user.name “Your Name”
• git config --global user.email yourname@email.com
• Verify that you have personalized the config file using the command cat ~/.gitconfig

```
queleroyubuntu3@workstation:~/CPE212_QUE$ git config --global user.name "quelero
yubuntu3"
queleroyubuntu3@workstation:~/CPE212_QUE$ git config --global user.email qjlpque
@tip.edu.ph
queleroyubuntu3@workstation:~/CPE212_QUE$ cat ~/.gitconfig
[user]
        name = queleroyubuntu3
        email = qjlpque@tip.edu.ph
```

h. Edit the README.md file using nano command. Provide any information on the markdown file pertaining to the repository you created. Make sure to write out or save the file and exit.

```
queleroyubuntu3@workstation:~/CPE212_QUE$ cat README.md
# CPE212_QUE
kahit ano
```

i. Use the git status command to display the state of the working directory and the staging area. This command shows which changes have been staged, which haven’t, and which files aren’t being tracked by Git. Status output does not show any information regarding the committed project history. What is the result of issuing this command?

```
queleroyubuntu3@workstation:~/CPE212_QUE$ git status
On branch main
Your branch is up to date with 'origin/main'.

Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
        modified:   README.md

no changes added to commit (use "git add" and/or "git commit -a")
```

Analysis: It shows that there are changes in the README.md

j. Use the command git add README.md to add the file into the staging Area.

```
queleroyubuntu3@workstation:~/CPE212_QUE$ git add README.md
```

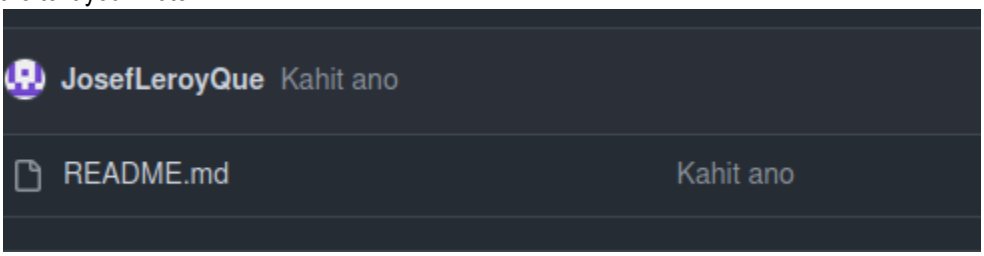
k. Use the git commit -m “your message” to create a snapshot of the staged changes along the timeline of the Git projects history. The use of this command is required to select the changes that will be staged for the next Commit.

```
queleroyubuntu3@workstation:~/CPE212_QUE$ git commit -m "Kahit ano"
[main 4a3e529] Kahit ano
 1 file changed, 2 insertions(+), 1 deletion(-)
```

l. Use the command git push <remote><branch> to upload the local repository content to GitHub repository. Pushing means to transfer commits from the local repository to the remote repository. As an example, you may issue git push origin main.

```
queleroyubuntu3@workstation:~/CPE212_QUE$ git push origin main
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Writing objects: 100% (3/3), 268 bytes | 268.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0
To github.com:JosefLeroyQue/CPE212_QUE.git
   ce0b814..4a3e529  main -> main
```

m. On the GitHub repository, verify that the changes have been made to README.md by refreshing the page. Describe the README.md file. You can notice the how long was the last commit. It should be some minutes ago and the message you typed on the git commit command should be there. Also, the README.md file should have been edited according to the text you wrote.



Analysis: Based on the texts I inserted, we can see how it proved that the commits were successful by seeing the “Kahit ano” text.

Reflections:
Answer the following:
3. What sort of things have we so far done to the remote servers using ansible Commands?
- We did the pinging of servers,running commands for both servers concurrently, and managing the servers easily through the workstation/local machine only.
4. How important is the inventory file?
- An inventory file is like storage for all the things that we need to manage. This would help the ansible know how to manage its servers, and know what commands to run.

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2. CONCLUSION / LEARNINGS

For this laboratory activity, I have learned how to do authentication without password permission by using the 4096-bit RSA SSH keys which connect to remote servers and link the local machine with GitHub. I have also learned how to do staging, committing and pushing changes to GitHub using only the local machine. Overall I have learned how the public-key cryptography enhances our system security and how the local Git workflows work with remote cloud hosting.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used: **N/A**

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: **N/A**

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use: **N/A**

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

By submitting this activity, I affirm that my use of AI complies with T.I.P.'s AI usage policy and does not exceed the permitted limits for similarity, automated content generation, or academic assistance.

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